

Diffusion Processes for Inference of Structural Causal Models

IA pour la Science

7 May 2026

1 Linear models excluding State-Space models

List of SCM-model parameters:

$\mathbf{w}$ the linear parameters of the model,

$\mathbf{v}$ the non-linear parameters of the model,

the covariance (mutual information) of the variables $X_1, \dots, X_n$,

1. the covariance of the linear part of the parameters $\mathbf{w}$,
2. the covariance of the non-linear part of the parameters $\mathbf{v}$.

The transformations

1. only G to f it admissible (there is no conversion from the model to G)

Architectures of the linear models:

1. linear model $\hat{\mathbf{x}} = \mathbf{W}^T \mathbf{x}$: a single matrix (2-index) that maps $X_1, \dots, X_n$ to $\hat{X}_1, \dots, \hat{X}_n$,
2. generalized linear model $\sigma \circ \mathbf{W}^T \mathbf{x}$: non-parametric link function of the linear model,
3. special non-linear model $g_1(\mathbf{v}_i) \circ g_r(\mathbf{v}_r)(\mathbf{x})$: a model with functions $g_i(\mathbf{v}_i)(X_j)$,
4. general non-linear model $\mathbf{W}^T g_i(\mathbf{v}_i)(X_j)$ (obsoleted): a linear combination of the model with functions,
5. generalized non-linear model $\sigma \circ \mathbf{W}^T g_i(\mathbf{v}_i)(X_j)$: a non-linear non-parametric transformation of the linear combination

Optimization:

1. for linear models is MLS,

2. for non-linear models is Levenberg-Marquardt,
3. for the non-linear models is Levenberg-Marquardt,
4. for the rest it will

Here the notation $\circ$ means the composition of functions. It could be replaced by brackets.

2 Ideal Magnetohydrodynamics (MHD) Equations

The equations

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0, \quad (1)$$

$$\frac{\partial(\rho \mathbf{v})}{\partial t} + \nabla \cdot (\rho \mathbf{v} \mathbf{v} - \mathbf{B} \mathbf{B}) + \nabla p = 0, \quad (2)$$

$$\frac{\partial \mathbf{B}}{\partial t} - \nabla \times (\mathbf{v} \times \mathbf{B}) = 0, \quad (3)$$

are a simplified form of the *ideal magnetohydrodynamics (MHD)* equations. They describe the dynamics of an electrically conducting fluid (plasma, ionized gas, liquid metal, etc.) interacting with a magnetic field.

The primary variables are

- $\rho(\mathbf{x}, t)$: mass density,
- $\mathbf{v}(\mathbf{x}, t)$: fluid velocity,
- $\mathbf{B}(\mathbf{x}, t)$: magnetic field,
- $p(\mathbf{x}, t)$: pressure.

2.1 Continuity Equation

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0. \quad (4)$$

This equation expresses conservation of mass.

The first term,

$$\frac{\partial \rho}{\partial t},$$

measures the rate of change of density at a fixed point in space.

The second term,

$$\nabla \cdot (\rho \mathbf{v}),$$

represents the divergence of the mass flux. It measures the net amount of mass flowing out of a small control volume.

If more mass leaves a region than enters it, the density decreases. If more mass enters than leaves, the density increases.

Thus the continuity equation states that mass can neither be created nor destroyed.

2.2 Momentum Equation

$$\frac{\partial(\rho \mathbf{v})}{\partial t} + \nabla \cdot (\rho \mathbf{v} \mathbf{v} - \mathbf{B} \mathbf{B}) + \nabla p = 0. \quad (5)$$

This equation is the fluid analogue of Newton's second law.

The quantity

$$\rho \mathbf{v}$$

is the momentum density of the fluid.

The term

$$\frac{\partial(\rho \mathbf{v})}{\partial t}$$

represents the local time rate of change of momentum.

The tensor

$$\rho \mathbf{v} \mathbf{v}$$

describes transport of momentum by fluid motion. It accounts for momentum carried from one region to another.

The tensor

$$\mathbf{B} \mathbf{B}$$

represents magnetic stresses. Magnetic field lines behave somewhat like stretched elastic strings: they exert tension along their length and pressure perpendicular to themselves.

The pressure-gradient term

$$\nabla p$$

describes the force exerted by pressure differences. Fluid tends to move from regions of higher pressure toward regions of lower pressure.

Overall, the momentum equation states that the fluid momentum changes because of pressure forces and magnetic forces.

2.3 Induction Equation

$$\frac{\partial \mathbf{B}}{\partial t} - \nabla \times (\mathbf{v} \times \mathbf{B}) = 0. \quad (6)$$

This equation governs the evolution of the magnetic field.
The quantity

$$\mathbf{v} \times \mathbf{B}$$

describes the interaction between fluid motion and the magnetic field. Its curl determines how the magnetic field changes in time.

A fundamental consequence of this equation in ideal MHD is the *frozen-in flux property*:

Magnetic field lines move with the fluid.

If the plasma stretches, twists, or compresses, the magnetic field lines stretch, twist, or compress with it.

2.4 Additional Equations

A complete ideal MHD system normally includes the constraint

$$\nabla \cdot \mathbf{B} = 0, \quad (7)$$

which expresses the absence of magnetic monopoles.

An equation of state or energy equation is also required to determine the pressure. For example,

$$p = (\gamma - 1)\rho e, \quad (8)$$

where e is the specific internal energy and γ is the adiabatic index.

Without such a relation the system is not closed.

3 Where is Space in the Equations?

The spatial coordinates are implicit in the notation.

More explicitly, each field depends on both position and time:

$$\rho = \rho(\mathbf{x}, t), \quad (9)$$

$$\mathbf{v} = \mathbf{v}(\mathbf{x}, t), \quad (10)$$

$$\mathbf{B} = \mathbf{B}(\mathbf{x}, t), \quad (11)$$

$$p = p(\mathbf{x}, t), \quad (12)$$

where

$$\mathbf{x} = (x, y, z). \quad (13)$$

For example, the continuity equation can be written explicitly as

$$\frac{\partial \rho(x, y, z, t)}{\partial t} + \nabla \cdot (\rho(x, y, z, t) \mathbf{v}(x, y, z, t)) = 0. \quad (14)$$

The dependence on (x, y, z, t) is usually omitted for brevity.

3.1 Gradient Operator

The gradient of a scalar field p is

$$\nabla p = \left(\frac{\partial p}{\partial x}, \frac{\partial p}{\partial y}, \frac{\partial p}{\partial z} \right). \quad (15)$$

It measures how rapidly the pressure changes from one point to another.

3.2 Divergence Operator

For a vector field

$$\mathbf{F} = (F_x, F_y, F_z),$$

the divergence is

$$\nabla \cdot \mathbf{F} = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z}. \quad (16)$$

It measures the net outflow from an infinitesimal volume.

For the mass flux,

$$\nabla \cdot (\rho \mathbf{v}) = \frac{\partial(\rho v_x)}{\partial x} + \frac{\partial(\rho v_y)}{\partial y} + \frac{\partial(\rho v_z)}{\partial z}. \quad (17)$$

3.3 Curl Operator

The curl of a vector field is

$$\nabla \times \mathbf{F} = \begin{pmatrix} \frac{\partial F_z}{\partial y} - \frac{\partial F_y}{\partial z} \\ \frac{\partial F_x}{\partial z} - \frac{\partial F_z}{\partial x} \\ \frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} \end{pmatrix}. \quad (18)$$

It measures the local rotational tendency of the field.

3.4 One-Dimensional Example

If the system varies only in the x direction, then

$$\rho = \rho(x, t), \quad v = v(x, t).$$

The continuity equation reduces to

$$\frac{\partial \rho}{\partial t} + \frac{\partial(\rho v)}{\partial x} = 0. \quad (19)$$

In this form the independent variables are explicit:

- x is the spatial coordinate,
- t is time.

The full MHD equations generalize this idea to three spatial dimensions by using the compact differential operators ∇ , $\nabla \cdot$, and $\nabla \times$.

Therefore, space enters the equations through the position vector

$$\mathbf{x} = (x, y, z),$$

and through the spatial derivatives acting on the fields. The variables ρ , $\mathbf{v}$, $\mathbf{B}$, and p are defined at every point in space and evolve in time according to the MHD equations.

Two aspects of the SCM inference were discussed the last week:

1. The inference avoids the Integer Linear Programming problem by replacing the probabilities of edges

$$\mathbf{\Gamma} \in [0, 1]^{n \times n} \mapsto \mathbf{Z} \in \{0, 1\}^{n \times n}.$$

for the adjacency of edges. *NB: it is solved below.*

2. The list of variables is not fixed. For the set of observable variables $X_1, \dots, X_n$ the model has to forecast the list of variables $X_1, \dots, X_n, \dots, X_{n'}$. By default, the value $n' = 0$. *NB: it is under consideration.*

The foundation model includes the elements of SCM selection:

- 1) maximizing likelihood of the DAG at each step,
- 2) do-intervention actions each step on the sampled node.

The foundation model for the SCM inference

The main type of the foundation model carries the (Riemannian) diffusion process.

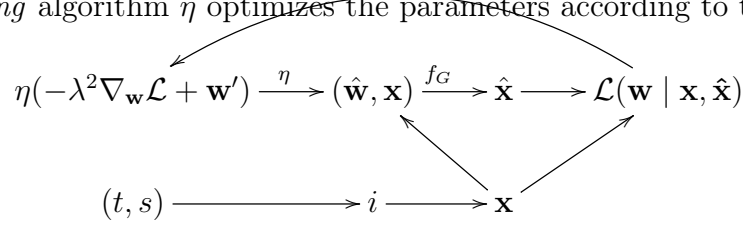
The diffusion process involves the step t and two distributions to approximate the distribution of the adjacency matrix $\mathbf{Z}$. Use the vector form $\mathbf{z} = \text{vec}(\mathbf{Z})$. Denote the training distribution $q(\mathbf{z})$ and the sampling distribution $p(\mathbf{z})$. During the model training one has to minimize the distance between two distributions at each step.

The diffusion process invokes the local structural modeling:

$$\mathbf{Z} \mapsto f_g \mapsto \eta \mapsto \mathbf{w},$$

with η is the optimization algorithm that delivers the optimal parameters to the SCM at each step of the diffusion process.

The *fine-tuning* algorithm η optimizes the parameters according to the diagram



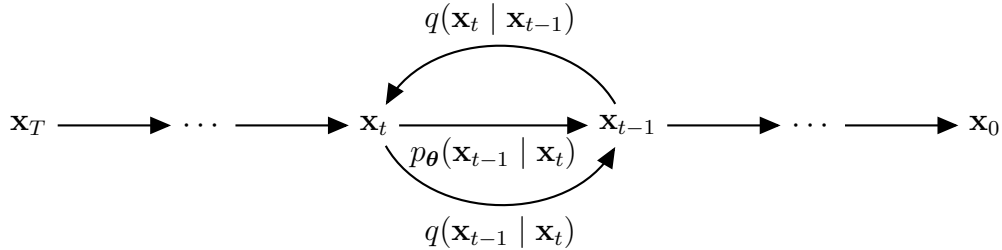
This diagram reads: the optimization algorithm η estimates the SCM parameters $\mathbf{w}$ by the gradient of its loss and the previous parameters to solve the problem

$$\hat{\mathbf{w}} = \arg \min \mathcal{L}(\mathbf{w}) \quad \text{given} \quad f_G(\hat{\mathbf{w}}, \mathbf{x}) \mapsto \hat{\mathbf{x}}.$$

The algorithm η uses some initial value $\mathbf{w}_0$ and a stop-criterion for its iterations.

A basic diffusion process for the data

The data $\mathbf{x}$ only, without the binary adjacent $\mathbf{z}$ for the model f_G are sampled and inferred with the generative process below.



Given the data distribution $\mathbf{x}_0 \sim q(\mathbf{x})$ assert:

- 1) forward diffusion process $\mathbf{x}_t = \sqrt{1 - \beta_t} \mathbf{x}_{t-1} + \sqrt{\beta_t} \mathbf{z}_t$, sampling i.i.d.

$$\mathbf{x}_1, \dots, \mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I}),$$

- 2) sampling $q(\mathbf{x}_{t-1} | \mathbf{x}_t)$ and learning the parameters $\boldsymbol{\theta}$ of f_{ai} : $p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1} | \mathbf{x}_t)$,

3) reverse diffusion process

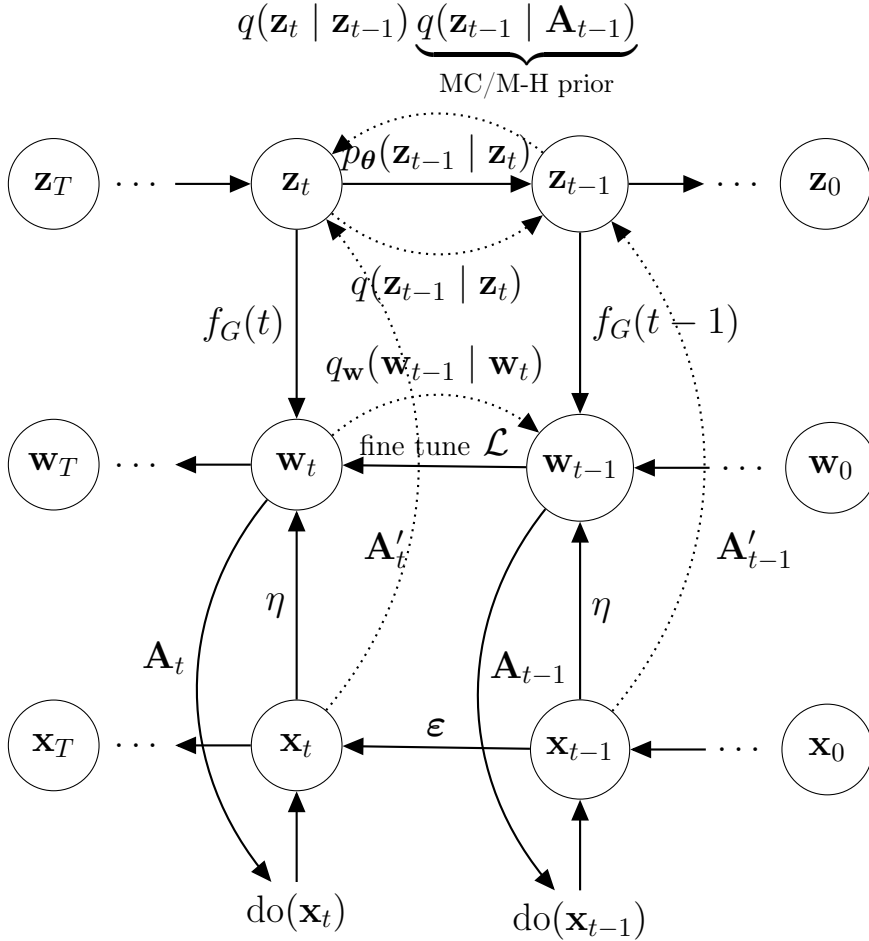
$$p_{\theta}(\mathbf{x}_{0:T}) = p(\mathbf{x}_T) \prod_{t=1}^T p_{\theta}(\mathbf{x}_{t-1} \mid \mathbf{x}_t),$$

4) slow learning gives

$$p_{\theta}(\mathbf{x}_{t-1} \mid \mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1}; \boldsymbol{\mu}_{\theta}(\mathbf{x}_t, t), \boldsymbol{\Sigma}_{\theta}(\mathbf{x}_t, t)).$$

Coherent generation of SCM submodels with do-intervention

Each step of the diffusion delivers a set of submodels to reconstruct. The dotted lines of the diagram are sampling; the rest are inference or optimization.



The algorithm of SCM inference includes the U-Net $f_{\text{ai}}(\boldsymbol{\theta})(\mathbf{x}, \mathbf{z})$ to learn its parameters $\boldsymbol{\theta}$ and make an inference of the SCM f_G along with its parameters $\mathbf{w}$. The estimated parameters $\hat{\mathbf{w}}$ are inherited at each step t . So that only fine-tuning optimization η goes to minimize $\mathcal{L}(\mathbf{w}_{t-1}) \mapsto \mathbf{w}_t$. (The dataset refers to the stricture $(\mathbf{X}, \mathbf{z})$ with a transformer.)

1. Sample a new SCM adjacency $\mathbf{z}_t$.
2. Construct the new model f_G and its initial parameters $\mathbf{w}_t$ from the step $t - 1$.
3. Fine-tune the SCM parameters $\hat{\mathbf{w}}_t = \arg \min \mathcal{L}(\mathbf{w}_t \mid \hat{\mathbf{w}}_{t-1})$ with the initial parameters.
4. Estimate the covairiance $\hat{\mathbf{A}}^{-1} = \mathbf{cov}(\mathbf{w})$ using the hessian of $\mathcal{L}$, the Fisher matrix.
5. Apply the CI-criterion $\rho(\mathbf{A}_t)$ to make the intervention $\text{do}(\mathbf{x}_t)$.
6. Make the intervention, sampling $\text{do}(\mathbf{x}_t)$ with the intervention parameters $\mathbf{A}'_t$
7. Compute the partial residuals $\mathbf{x}_{t-1} - \hat{\mathbf{x}}_{t-1} = \boldsymbol{\varepsilon}_t \mapsto \mathbf{x}_t$.
8. Learn the f_{ai} parameters by reconstructing $p_{\boldsymbol{\theta}}(\mathbf{z}_{t-1} \mid \mathbf{z}_t)$.
9. Go to the step 1 and sample the next adjacency $\mathbf{z}_{t-1}$.

List of the modules

The empirical distribution of the observations

$P(X_1, \dots, X_n)$ is the distribution of the observable data. Its approximation is given by the Gaussian Process Regression model

$$p(\mathbf{x} \mid f_{\text{gp}})$$

on the dataset

$$\{\mathbf{x}_i \in \mathbb{R}^n\}.$$

It is indexed by some function of the time and space $(t, s) \mapsto i$.

A model from the family of universal approximators

For a sample $\mathbf{x}$ and its reconstruction $\hat{\mathbf{x}}$ the likelihood function provided a universal model $f_{nn}(\boldsymbol{\theta}, \mathbf{x})$ is

$$p(\mathbf{x} \mid f_{nn}, \boldsymbol{\theta}).$$

Without the a priori information about the model parameters $\boldsymbol{\theta}$ the corresponding loss function is

$$\mathcal{L}(\boldsymbol{\theta}) = -\log p(\mathbf{x} \mid f_{nn}, \boldsymbol{\theta}) = (\hat{\mathbf{x}} - \mathbf{x})^\top \mathbf{B}^{-1}(\hat{\mathbf{x}} - \mathbf{x}).$$

The matrix $\mathbf{B}$ is the covariance of the reconstruction error. This matrix in some states depends on the index i . This model as a parametric one has a structure to make a distillation to the causal graph

$$f_{nn}(\boldsymbol{\theta}, \mathbf{x}) \xrightarrow{\text{extract}} \mathbf{Z}_{nn} \xrightarrow{\text{dist}} \mathbf{Z}.$$

An SCM we have to forecast

A structural causal model

$$\hat{\mathbf{x}} = f_G(\mathbf{w}, \mathbf{x}) \in \mathcal{F}_{\text{scn}}$$

has a structure $\mathbf{Z}$ and its probabilistic representation $\boldsymbol{\Gamma}$. The elements of the matrix $\boldsymbol{\Gamma}$ are the probabilities of the causal relationships between the elements of the model structure f_G . The matrix $\mathbf{Z}$ is the adjacency matrix of the causal graph, the binary representation of $\boldsymbol{\Gamma}$.

Note: we suppose that SCM does not fit the whole (s, t) -data but a local part of it. So we also call SCM a local model f_G , in contrast to the universal model f_{nn} .

A foundational model to forecast the SCM

There given pairs a sample $\mathbf{x}_i$ and the corresponding SCM f_G with its structure $\{\mathbf{Z}_i\}$. Call the set of pairs

$$(\mathbf{x}_i, \mathbf{Z}_i), \quad i \in \mathcal{B}$$

a collection of the SCM forecasting. This collection $\mathcal{B}$ shows one case of mathematical modeling that results in an interpretable model f_G . This model fits samples $\mathbf{x}_i$ from some local data $i \in \mathcal{B}$.

Call a model that forecasts an SCM $\mathcal{F}$ a *foundation model*

$$f_{\text{ai}} : \mathbf{x}_{\mathcal{B}} \mapsto f_G \in \mathcal{F}.$$

The SCM is a parametric model

$$f_G = f_G(\mathbf{w}, \mathbf{x}) \quad \text{with the parameters} \quad \hat{\mathbf{w}} = \arg \min_{\mathbf{w}} \mathcal{L}(\mathbf{w}).$$

The foundational model returns not the SCM $f_G(\hat{\mathbf{w}}, \mathbf{x})$ itself, but only a probabilities $\boldsymbol{\Gamma}$ on the structure $\mathbf{Z}$. So we rewrite the previous definition as

$$f_{\text{ai}} : \mathbf{x}_{\mathcal{B}} \mapsto \boldsymbol{\Gamma}.$$

Note: Since a case of mathematical modeling, call it $\mathcal{B}$ has a history of requests to the foundational model, the next information is available for it:

- 1) the optimal parameters $\mathbf{w}$ along with $\mathbf{Z}, \mathbf{\Gamma}$ for the SCM f_G ,
- 2) the prior probabilities of the model parameters $p(f_G)$ or all models in the family $\mathcal{F}$,
- 3) the other information about the case of mathematical modeling $\mathcal{B}$ that is relevant for the forecasting of the SCM.

The Integer Linear Programming for the causal graph selection

The matrix $\mathbf{\Gamma}$ comes along with the descriptions of its nodes $X_1, \dots, X_n$, and the probabilities. The SCM f_G is defined by the graph $G = (\mathbf{Z}, X)$ by solving

$$\text{ILP}(\mathbf{\Gamma}, P_X) \rightarrow \mathbf{Z} \rightarrow f_G.$$

The evidence of the SCM

A probabilistic description $\mathbf{\Gamma}$ of a model f_G from the class of *admissible models* $\mathcal{F}$ is generated from a parametric distribution

$$\mathbf{\Gamma} \sim \text{Dir}(\boldsymbol{\pi}) \quad \text{generated by MCMC} \quad f_{\text{mh}}(\boldsymbol{\pi}),$$

here the lowercase letters mean Metropolis-Hastings Algorithm for MCMC. (A simple genetic algorithm may bring some decent results, too). This distribution provides a probabilistic space for the similar models. See the document on *Multimodeling*.

The model f_G and the corresponding graph has its likelihood

$$p(\mathbf{x}|f_G).$$

See the document on *Causal graph likelihood*.

Note: *Two-level bayesian inference introduces the probability of the SCM parameters on the first level. And the probability of the model on the second level is:*

- 1) the prior of the SCM is delivered by the generative model f_{mh}
- 2) the likelihood of the SCM is delivered by the forecasting model f_{ai} .

The criterion of causality

The parameters of the model f_G has their covariance matrix. The prior of the model parameters is

$$-\ln p(\mathbf{w}|f_G) = (\mathbf{w} - \boldsymbol{\mu})^\top \mathbf{A}^{-1}(\mathbf{w} - \boldsymbol{\mu}).$$

The algorithm of the Bayesian hyperparameters $(\boldsymbol{\mu}, \mathbf{A})$ estimation (and their connection between the causality) is described in the document on *parameter analysis*.

A model for the intervention

See the document on the *intervention strategy*.